

AYMAN MOSTAFA

+20 115 865 8380 ◊ Maadi, Cairo, Egypt

ayman.msayed1@gmail.com ◊ [LinkedIn](#) ◊ [GitHub](#) ◊ [CodeForces](#)

EDUCATION

Bachelor's degree in Computer Science, Ain Shams University October/2022 - June/2026 (Expected)

Year: Senior | **Department:** Computer Science

Relevant Coursework: Algorithms, Data Structures, Artificial Intelligence, Object-Oriented-Programming, Databases, Operating Systems, Computer Architecture, System Analysis and Design, Computer Networks.

SKILLS

Programming Languages C++, C, C#, HTML, CSS, Java, Python, JavaScript, Scala

Technical Skills Git, GitHub, OOP, Databases, Linux, NodeJS, React

Soft Skills Leadership, Planning, Teamwork, Troubleshooting, Research, Interest and Ability to Learn Fast

Languages Arabic (Native), English (Proficient)

PROJECTS

Valorithms – Image Segmentation (C#, Windows Forms) ([Repo Link](#))

- Implemented Felzenszwalb image segmentation with brute-force and optimized 5D kd-tree neighborhood search, achieving a **+10% improvement in segmentation accuracy**.
- Designed graph construction functions using both brute-force methods and kd-tree-based optimization for efficient neighborhood computation.
- Built a custom Disjoint Set Union (DSU) structure to apply Minimum Spanning Tree (MST)-based region merging during segmentation.
- Ranked **1st place among 120+ teams** in final evaluation based on segmentation performance and accuracy.

FOS – A Simple Operating System (C, Assembly) ([Repo Link](#))

- Implemented kernel and user heap memory allocation using first-fit strategy, increasing allocation speed by **85%** and reducing memory usage by **25%**.
- Developed an efficient priority-based task scheduler, improving task throughput by **40%** and reducing average wait time by **20%**.
- Implemented shared memory object management, enabling inter-process communication and improving resource-sharing efficiency.
- Ranked **4th place out of 125+ teams** in final evaluation based on implementation quality and performance.

C-DS Data Structure Visualizer (C++, ImGUI) ([Repo Link](#))

- Developed interactive, real-time visualization modules for data structures and algorithms, improving newcomer learning efficiency by **40%**.
- Designed visualizations for core data structures (stack, queue, deque, sparse table) with animated, step-by-step interaction to enhance clarity and conceptual understanding.

Dr Stone: Civilization Rebooted (C++, SFML) ([Repo Link](#))

- Collaborated in a team project awarded **1st place** in the Freshman Best Projects for quality and innovation.
- Created animations and hitboxes with **95% accuracy** across **10+ targets**, boosting responsiveness by **30%**.
- Integrated graph algorithms for pathfinding and custom data structures, optimizing real-time game updates by **25%**.

EXTRA-CURRICULAR ACTIVITIES

- **Mentor** – [acmASCIS Student Activity](#) Mentored **30+** **trainees** in algorithms and data structures for competitive programming. Designed, set, and tested **20+** **problems** used in training sessions and local contests.
- **Head of Human Resources** – [Potentia](#) Led recruitment and internal communications, strengthening team collaboration and cohesion. Organized student events and midyear workshops, enhancing member engagement and skills development.

ACHIEVEMENTS

- Ranked **125th** in the ICPC ECPC 2024 (Egyptian Collegiate Programming Contest).
- Ranked **54th** out of 300+ teams in the ECPC 2024 Qualifiers, qualified for the main event.
- Ranked **77th** out of 250+ teams in the ECPC 2023 Qualifiers.
- Won **1st place** in the IBM Structured Programming Best Projects Award 2023 at Ain Shams University.
- Solved **1500+** **coding problems** on online judges and reached **Specialist** rank on Codeforces.